

CURRENT AND PENDING (OTHER) SUPPORT

NSF Common Form — November 1, 2023 / DARPA-PA-26-05 MATHBAC Submission

Identifying Information

***Name:** Davis, Issac D.

Persistent Identifier (ORCID/PID): N/A

***Position Title:** Principal Investigator / Independent AI Safety Researcher

Organization and Location

***Organization Name:** ISSAC D DAVIS / SCBE-AETHERMOORE

***Location (City, State, Country):** Port Angeles, WA, USA

a. Proposals and Active Projects

Disclosure covers all proposals and active projects in accordance with NSPM-33 Current and Pending (Other) Support definitions. Two proposals are currently pending. No active projects with existing federal awards.

Project/Proposal 1

***Title:** SCBE Mathematical Framework for Agentic Communication Protocols

***Status of Support:** Pending (this proposal)

Award Number (if available): N/A — proposal not yet awarded

***Source of Support:** Defense Advanced Research Projects Agency (DARPA), Defense Sciences Office (DSO), MATHBAC Program. Solicitation: DARPA-PA-26-05.

***Primary Place of Performance:** Port Angeles, WA, USA

***Start Date (MM/YYYY):** 10/2026 (estimated; 16 months from anticipated award)

***End Date (MM/YYYY):** 01/2028 (estimated)

***Total Anticipated Amount:** \$839,000 (full proposal amount; inclusive of G&A and fee)

***Person-Months Per Year Devoted to Project:** 10 person-months per year (sole PI, full-time effort). Year 1: 10 months (months 1–12). Year 2: 4 months (months 13–16).

***Overall Objectives:** Develop a composed-operator governance framework $T = L14 \circ \dots \circ L1$ for multi-agent AI communication using hyperbolic geometry (Poincare ball model). Map agent communication to Poincare ball coordinates; enforce five physical axioms (Unitarity, Locality, Causality, Symmetry, Composition) layer-by-layer. Demonstrate exponential adversarial cost scaling via harmonic wall $H(d, pd) = 1/(1 + \phi * d_H + 2 * pd)$. Validate on NMR spectroscopy subdomains (Task A: 1H NMR to structure; Task B: 2D COSY assignment) with Karplus equation recovery as primary falsifiability experiment. Target: 99% adversarial classification, p95 latency <100ms for 240 cases.

***Statement of Potential Overlap:** This entry IS the MATHBAC proposal submission itself. A concurrent pending proposal (CLARA, DARPA-PA-25-07-02) uses the same SCBE governance

framework in a different task domain (theorem-proving / course-of-action planning) with no scientific overlap to the NMR spectroscopy scope proposed here. In the event both proposals are awarded, the PI will disclose person-month commitments to both Program Managers and adjust effort allocations to eliminate any budgetary duplication. Person-month overlap would require renegotiation of at least one award's period of performance.

Project/Proposal 2

***Title:** SCBE-AETHERMOORE: Verifiable Defeasible LP Composition for Trustworthy Multi-Agent AI Governance

***Status of Support:** Pending (submitted; award decision anticipated 06/2026)

Award Number (if available): DARPA-PA-25-07-02-CLARA-FP-033 (submission ID confirmed 2026-04-13)

***Source of Support:** Defense Advanced Research Projects Agency (DARPA), Defense Sciences Office (DSO), CLARA Program. Solicitation: DARPA-PA-25-07-02.

***Primary Place of Performance:** Port Angeles, WA, USA

***Start Date (MM/YYYY):** 06/2026 (if awarded; Phase 1 = 15 months, Phase 2 = 9 months)

***End Date (MM/YYYY):** 06/2028 (full 24-month program if both phases awarded)

***Total Anticipated Amount:** \$2,000,000 total (\$1,350,000 Phase 1 / \$650,000 Phase 2, approximate)

***Person-Months Per Year Devoted to Project:** 10 person-months per year (sole PI, full-time effort Phase 1). Phase 2 estimated 9 person-months per year.

***Overall Objectives:** Develop verifiable defeasible logic program (DLP) composition approaches for trustworthy multi-agent AI governance in complex systems engineering. SCBE's 14-layer pipeline provides the compositional AR+ML substrate. TA1 target: create composition approaches that achieve polynomial-time verifiability without performance loss, composed-task reliability exceeding state-of-the-art baselines, and sample complexity below SOA benchmarks (Phase 2). Application domain: cyber defense and/or COA planning (DARPA CLARA TA1 scope).

***Statement of Potential Overlap:** CLARA and MATHBAC both leverage the SCBE governance framework but address distinct task domains: CLARA targets theorem-proving / COA planning (DARPA TA1 compositional AR+ML); MATHBAC targets NMR spectroscopy (DARPA TA1 multi-agent science subdomains). There is no scientific overlap in objectives, datasets, or validation experiments. The shared technical substrate (SCBE framework) constitutes Background IP disclosed in both proposals. If both awards are made simultaneously, person-month commitments will be disclosed and adjusted as required by both Program Managers to eliminate any budgetary duplication.

b. In-Kind Contributions

None. No in-kind contributions with an estimated value of \$5,000 or more that require a commitment of the PI's time are currently received or pending.

Certification

I certify that the information provided is current, accurate, and complete. This includes, but is not limited to, information related to current, pending, and other support (both foreign and domestic) as defined in 42 U.S.C. §6605.

I also certify that, at the time of submission, I am not a party in a malign foreign talent recruitment program.

Misrepresentations and/or omissions may be subject to prosecution and liability pursuant to, but not limited to, 18 U.S.C. §§287, 1001, 1031 and 31 U.S.C. §§3729-3733 and 3802.

Signature: [Sign before submission]

Date: [Date — must be within 12 months of submission]

DRAFT — 2026-06-01. Prepared per NSF NSPM-33 Common Form (November 2023). OMB Control No. 3145-0279. Public reporting burden estimated 2 hours.